

RepoScope

The Graph Operating System for AI Coding Agents & Engineering Teams

CORE VALUE PROPOSITION (v2.8)
Replaces flat directory file scans with an in-place, typed AST code graph (.reposcope/graph.json) computed on localhost with 100% zero-upload privacy.

FOUR SYNCHRONIZED INGRESS SURFACES
1. Interactive Web Canvas (React Flow + D3 MindMap V2)
2. Headless Terminal CLI (index, run, mcp, security, blast)
3. Native VS Code / Cursor Extension + 4. 9-Agent MCP Harness

[v2.8.0 Unified Production Release]
• [100% Zero-Upload Local Privacy]

[9-Agent MCP Configurator]
Claude Code • Cursor • Codex • OpenCode • Windsurf • Gemini • Roo • Copilot • Cline

[6-Engine Contextual Security]
AST Taint Flow • Secrets • CVEs • Config Drift • 1-Click AST Patching

[Symbol-Level PR Safety]
15-Point Multi-Hop Blast Simulator • Zero File Caps • Exact Contract Breaks

GITHUB REPOSITORY
github.com/adityapatel5912/RepoScope

OPEN VSX EXTENSION REGISTRY
open-vsx.org/extension/adityapatel5912/reposcope

Every Repository is a Living Graph. We Still Read It as a Pile of Files.

Both human engineers and AI coding agents suffer when architectural truth lives in scattered, flat directories.

The Agent Context Tax (92% Waste)

- **Blind Directory Scanning:** LLMs waste precious context tokens reading irrelevant files.
- **Severe Hallucination Rate:** Without verified AST coordinates, agents invent broken APIs and imports.
- **Divergent Mental Models:** Agents miss module boundaries, causing monolithic architectural drift.

`[ ALERT: 92% Context Window Waste ]`

Invisible Multi-Hop Blast Radius

- **Human Grep Guesswork:** Developers guess downstream impact before merging PRs.
- **Silent Contract Breaks:** Renaming a parameter breaks callers 3 hops away with zero compiler warning.
- **Missing Test Coverage:** PRs modify high-centrality files while leaving downstream consumers untested.

`[ WARN: Invisible 3-Hop Regressions ]`

Tribal Onboarding & Knowledge Loss

- **Months to Full Productivity:** New hires spend weeks orienting without a topological map.
- **Architectural Drift:** System structure lives only in senior heads; documentation rots quickly.
- **Zero Shared Truth:** No single foundation unifies IDEs, CLI pipelines, and AI agents.

`[ LAG: Months to Orient New Engineers ]`

One In-Place Code Graph. Four Synchronized Ingress Surfaces.

Every surface reads .reposcope/graph.json so human developers and AI coding agents quote identical answers.

SHARED GRAPH OPERATING SYSTEM SPINE: Source ASTs → .reposcope/graph.json → Canvas • CLI • VS Code • 9-Agent MCP

Pillar 1: Multi-Language AST Parser

• **Tree-sitter Grammars:** In-process AST parsing for Python, TypeScript, TSX, JS, Go, and Rust.

• **Typed Entity Extraction:** Exact line coordinates for classes, functions, and routes.

• **Directed Graph:** Computes imports, calls, inheritance, containment, and taint propagation edges.

[6 Tree-sitter Grammars Supported]

Pillar 2: Zero-Upload Local Engine

• **Localhost Privacy:** reposcope run . indexes directly in place (127.0.0.1:4100) with zero telemetry.

• **SQLite & JSON Store:** Sub-second warm restarts with persistent graph store in .reposcope/graph.json.

• **Browser WASM Engine:** Tree-sitter WASM via File System Access API without installing a backend.

[100% Localhost Zero-Upload Privacy]

Pillar 3: Four Synchronized Surfaces

• **Interactive Web Canvas:** React Flow + D3 graph with layer badges and inspection drawers.

• **Headless CLI Tooling:** Terminal CLI with CI/CD exit codes and JSON output.

• **Universal MCP & VS Code:** 8-tool MCP server for agents + live status bar risk in the IDE.

[Canvas • CLI • VS Code • MCP]

Read the Macro Architecture Before Reading Individual Files.

Switch seamlessly between React Flow architectural layers and D3 hierarchical semantic mindmaps.

1. Connectivity Pyramid & Layers

• **Deterministic Layering:** Root → Backbone → Core → Services → UI → Config → Docs.

• **Color-Coded Layer Badges:** Nodes display visual role badges (Entrypoint, Core, Route, Utility).

• **Directory Grouping:** Toggle boundary boxes to inspect cross-module coupling.

• **Hierarchy Flow:** L1 Root ▸ L2 Backbone ▸ L3 Core ▸ L4 Services ▸ L5 UI ▸ L6 Config ▸ L7 Docs.

`[ 7 Hierarchical Layer Badges ]`

2. D3 Semantic MindMap V2

• **Dual Layouts:** One-click toggle between Left-to-Right (LR) tree flow and Top-to-Bottom (TB) hierarchy.

• **Interactive Folding:** Collapse or expand subtrees smoothly with fluid SVG camera animations.

• **Pure SVG Rendering:** Accessible, lightweight rendering ensures instant responsiveness on large repos.

`[ Left-to-Right & Top-to-Bottom Flow ]`

3. Real-Time Node Inspector

• **PageRank Hubs:** Mathematical importance and coupling degree of every module.

• **Direct Callers:** Side drawer displaying upstream dependents and exported symbols.

• **IDE Deep-Linking:** Click any symbol to trigger reposcope://open directly into VS Code / Cursor.

`[ PageRank Centrality & Symbol Links ]`

The Same Graph, Available to 9 Major Coding Agent Ecosystems.

Standardizing agent repository intelligence across Claude Code, Cursor, Codex, OpenCode, and Windsurf.

1. Supported Agent Ecosystems (9 Universal Agent Harnesses)

- **Claude Code & OpenCode:** Full stdio and Streamable HTTP support for Anthropic's official CLI.
- **Cursor & Codex / Windsurf:** Seamless integration via .cursor/mcp.json and .codecompanion.json.
- **Gemini CLI, Copilot, Roo Code, Cline:** Universal generator applies tailored JSON to all 9 agent setups.

`[ 9 Universal Agent Harnesses ]`

2. 8 High-Performance MCP Tools

- **get_architecture_overview:** Module boundaries, rules, entry points & stack.
- **calculate_blast_radius & get_dependency_lineage:** 3-hop impact & call chains.
- **get_repo_map & get_hotspots:** PageRank hierarchy and Git churn heatmaps.
- find_dependency_cycles, find_unreachable_symbols & explain_architectural_rule.

`[ 8 Query & Diagnostic Tools ]`

3. Dual Protocol & Safe Configurator

- **Dual Transport Support:** Runs over local stdio pipe or persistent Streamable HTTP (port 8000).
- **Eliminates SSE Dropouts:** Streamable HTTP avoids Server-Sent Event timeouts during heavy load.
- **One-Click Dialog:** Interactive modal allows developers to preview, copy, or write agent configs.

`[ Stdio & Streamable HTTP Protocol ]`

Predict Change Blast Radius and Catch PR Contract Breaks.

Stop guessing if a refactor breaks downstream callers. Simulate multi-hop impact before opening a PR.

1. Multi-Hop Blast Simulator

- **3-Hop Tracing:** Direct callers (Hop 1), indirect dependents (Hop 2), API bounds (Hop 3).
- **15-Point Risk Scoring:** Mathematical risk calculation weighing centrality and critical paths.
- **Visual Risk Badges:** Critical (8/10), High (7/10), and Moderate badges warn of breaks.

`[ 15-Point Multi-Hop Risk Scoring ]`

2. Symbol-Level PR Analyzer

- **Zero File Caps:** Analyzes complete git diffs across any number of changed files.
- **Exact Contract Breaks:** Pinpoints renamed parameters, modified returns, and altered routes.
- **Remediation Hints:** Suggests exact downstream call-sites that require updates.

`[ Zero File Limits • Exact Contracts ]`

3. Test Coverage Gap Alerts

- **Uncovered Dependent Alerts:** Cross-references affected downstream nodes against test graph.
- **GitHub CI/CD Bot:** CLI generates rich Markdown and Mermaid impact diagrams on PR comments.
- **Pre-Commit Defense:** Run reposcope blast <file> locally before pushing to CI.

`[ GitHub CI/CD Action Comments ]`

Graph-Contextual Security: From Input Taint to Dangerous Sinks.

Six specialized vulnerability engines paired with automated AST syntax patch generation.

1. AST Taint Flow & Injections

• **Source-to-Sink Tracing:** Traces request.args through functions directly into sinks.
• **Dangerous Sinks:** Flags raw SQL execution (cursor.execute), command injection, and eval.
• **Contextual Trace Tree:** Visual path from HTTP entry to database execution.

`[ Source-to-Sink Taint Propagation ]`

2. Secrets, CVEs & Config Drift

• **Zero False-Positive Secrets:** Detects API keys, JWT tokens, and leaked .env files.
• **Graph-Aware CVEs:** Maps dependency advisories to actual AST call-sites.
• **Auth Bypass & Config Drift:** Flags unauthenticated administrative routes and permissive CORS.

`[ 0 False-Positive Redaction ]`

3. Automated AST Patching

• **Syntax Patch Viewer:** Generates verified, syntactically correct AST replacement patches.
• **Parameterized Queries:** Transforms concatenated SQL into safe parameterized calls.
• **One-Click CLI Fix:** Run reposcope security --fix to apply patches to the working tree.

`[ 1-Click Syntactic Code Patches ]`

AST SYNTACTIC PATCH & CLI REMEDIATION

```
- cursor.execute(f"SELECT * FROM users WHERE id = {uid}")
+ cursor.execute("SELECT * FROM users WHERE id = ?",
(uid,))
$ reposcope security --fix
```

Turn Complex Codebases into Structured, Topological Learning Paths.

Kahn DAG curriculum tiers and interactive guided tours replace scattered wiki pages and tribal meetings.

1. Topological Curriculum (Kahn DAG)

- **Tier L1 (Core Foundations):** Zero-dependency base utilities, data models, and schemas.
- **Tier L2 (Domain Services):** Business logic engines and repository data pipelines.
- **Tier L3 (APIs & Routes):** HTTP controllers, REST routes, and auth middleware.
- **Tier L4 (Integrations & UI):** Frontend views, external APIs, and CLI commands.

`[ L1–L4 Kahn DAG Topological Sort ]`

2. Interactive Architectural Katas

- **'Why It Matters' Notes:** Human-readable architectural role summaries for every module.
- **Checkpoint Quizzes:** Interactive challenges test understanding of data flows.
- **Canvas Highlighting:** Selecting tiers illuminates the exact path on the graph.

`[ 'Why It Matters' Knowledge Quizzes ]`

3. 4 Deterministic CodeTours

- **System Overview Tour:** 7-step orientation guiding developers through entry points.
- **Security & Auth Track:** Follows request authentication to permission checks.
- **Request Lifecycle:** Traces HTTP requests from routing to database commit.
- **VS Code .tour Export:** Exports guided tracks to standard .tour files for IDE playback.

`[ Native VS Code .tour Playback ]`

Compile Real Repository Architecture into Structured LLM Prompts.

Deliver pristine, token-optimized context packs directly into Claude, ChatGPT, Cursor, and Gemini.

TOKEN OPTIMIZATION ENGINE: 50,000-Token Raw Dump ⟶ 3,000-Token 2-Hop Closure Blueprint (85–92% Token Savings)

1. Context & Feature Slice Modes

- **Context Mode:** Synthesizes overview, module boundaries, and conventions into an executive prompt.
- **Feature Slice Mode:** Extracts target feature plus its exact 2-hop interface closure.
- **85–92% Token Savings:** Replaces massive 50k dumps with 3k-token context blueprints.

`[ 2-Hop Closure • 85–92% Token Savings ]`

2. Rebuild Spec & Test Suite Modes

- **Rebuild Spec Mode:** Generates specifications, schemas, and API contracts to re-implement modules.
- **Test Suite Mode:** Analyzes call hierarchies to construct targeted test suites.
- **Contract Assertions:** Prompts include AST assertions enforcing strict types.

`[ Contract-First Rebuild & Test Specs ]`

3. Interactive Compiler UI & CLI

- **Modal Selector:** Select target nodes, toggle mode, and preview markdown in real-time.
- **Clipboard Copy:** One click copies token-counted blueprint directly to clipboard.
- **Headless CLI:** Run reposcope prompt --mode feature --target <symbol>.

`[ Instant Clipboard Copy & CLI Engine ]`

Universal Assistant: Zero-Hallucination Q&A with AST Citations.

Ask questions about your codebase. Get streaming answers backed by clickable, verified AST citations.

1. AST-Verified Citation Engine

- **Zero-Hallucination Guarantee:** Code quotes are cross-referenced with .reposcope/graph.json.
- **Green Pill Badges:** Verified references render with a green badge and exact coordinates.
- **Interactive Snapping:** Clicking citations focuses the node and opens the file in VS Code.

`[ 0% Hallucination Guarantee ]`

2. Universal Action Cards

- **Embedded UI Cards:** Assistant responses include actionable UI buttons in the stream.
- **[Launch Tour] & [Run Security Audit]:** Begin guided walkthroughs or trigger scans instantly.
- **[Simulate Blast Radius]:** Simulates impact of suggested refactorings directly from chat.

`[ Contextual Action Cards in Stream ]`

3. BYOK Multi-Model Flexibility

- **Bring Your Own Key:** Connect Google Gemini, Anthropic Claude, OpenAI, or DeepSeek.
- **100% Local LLM:** Route queries to local Ollama / LM Studio endpoints.
- **Streaming SSE:** Fluid token-by-token delivery with low latency.

`[ Gemini • Claude • OpenAI • Ollama ]`

VERIFIED CITATION & ACTION STREAM

```
Citation: [✓ backend/graph_builder.py:142 (build_graph)]
Citation: [✓ backend/security.py:88 (trace_taint_flow)]
Actions:  [Launch Tour]  [Run Security Audit]  [Simulate Blast Radius]
```

`[ reposcope://open?file=backend/graph_builder.py&line=142 ]`

Executable Scaffolding and Deep Repository Health Telemetry.

Spin up clean architectural starter kits in seconds and identify high-risk co-change hotspots.

1. Clean Architecture Scaffolder

- **Executable Starter Kits:** Multi-file scaffolds for FastAPI, React, and Go.
- **Hexagonal Clean Patterns:** Enforces domain isolation, repository interfaces, and test mocks.
- **Interactive Preview:** Inspect generated file hierarchies and code in the UI.
- **CLI Execution:** Copy-paste one-line CLI command to scaffold new modules.

`[ Executable Multi-File Scaffolds ]`

2. Git Churn & Co-Change Hotspots

- **Commit Churn:** Analyzes git history (HEAD~10..HEAD) against AST complexity.
- **Co-Change Coupling:** Discovers hidden temporal dependencies between files.
- **Drift Warnings:** Warns when frequently changed files gain excessive dependencies.

`[ Git Diff Churn & Coupling Matrix ]`

3. Engineering Health Metrics

- **Bus Factor:** Identifies critical modules maintained by single contributors.
- **Cycle & Dead Code:** Calculates Tarjan SCC cycles and finds unreachable symbols.
- **PageRank Hubs:** Ranks the top 10 most influential backbone components.

`[ PageRank Hubs & Bus Factor ]`

The Graph Follows Your Editor. 100% Local-First Privacy.

Real-time IDE intelligence without sending a single line of proprietary code to external clouds.

1. Native VS Code / Cursor Extension (open-vsx.org/extension/adityapatel5912/reposcope)

- **Live Status Bar:** Real-time dependent count and risk level for active file.
- **CodeLens Callers:** Click 'View Callers (6)' directly above exported functions.
- **Activity Bar Mini-Graph:** Renders 2-hop neighborhood around open file.
- **Open VSX Registry:** Packaged and distributed on Open VSX (adityapatel5912.reposcope).

`[ Live Status Bar & CodeLens Callers ]`

2. Zero-Upload Local Privacy

- **Localhost Architecture:** reposcope run . executes 100% on localhost (127.0.0.1:4100).
- **Air-Gapped Compliance:** Run safely on sensitive enterprise codebases with zero leakage.
- **In-Memory AST Caching:** Sub-millisecond graph queries served from local memory.

`[ 100% Localhost Zero Telemetry ]`

3. Browser-Local WASM Engine

- **showDirectoryPicker API:** Open repository folders in Chrome/Edge directly.
- **Tree-sitter WASM:** Parses code directly inside the browser sandbox: 75+ files in <2s.
- **Zero-Install Demos:** Test RepoScope instantly on sample repos without terminal setup.

`[ In-Browser Tree-sitter WASM Engine ]`

Empirical Evidence: 92% Token Reduction and 4.3× Faster Tasks.

Rigorous automated testing and benchmark validation across speed, accuracy, and token efficiency.

92% — Agent Token Savings

4.3× — Faster Task Completion

335+ — Verified Automated Tests

Agent Token Savings

- **45k → 3.6k Tokens:** Feature Slice reverse-prompts eliminate 92% of redundant context tokens.
- **$1,400+ Cost Savings:** Massive token reduction drastically cuts LLM API spend across teams.
- **Eliminates Context Drift:** Focusing strictly on 2-hop closures prevents hallucinations.

`[ 45,000 → 3,600 Tokens (92% Cut) ]`

Faster Task Completion

- **Sub-Second Navigation:** calculate_blast_radius and get_repo_map target symbols instantly.
- **12.8s vs. 54.2s Bug Fixes:** Agents fix multi-file bugs 4.3× faster than with flat search.
- **Zero Broken PR Merges:** Contract break checks catch 100% of breaking changes.

`[ 12.8s vs. 54.2s Multi-File Resolution ]`

Verified Automated Tests

- **Pytest Test Suite:** 335 passing backend unit, integration, and security tests.
- **Playwright E2E Suite:** End-to-end browser tests verifying graph at 2x Retina DPI.
- **Source-Available 1.0:** Free to inspect, customize, and self-host for internal teams.

`[ 335+ Pytest & Playwright Tests Pass ]`

Get Started with RepoScope in 30 Seconds.

Turn your codebase into a living, queryable code graph today.

1. QUICKSTART INSTALL	2. CONNECT YOUR AGENTS	3. JOIN THE COMMUNITY

`$ pip install reposcope`
`$ reposcope run .`
`# Or via npm:`
`$ npx @reposcope/cli run .`
	`$ reposcope mcp . --apply-all`
	
	`# Auto-configures 9 agents:`
	`# Claude, Cursor, OpenCode, Codex...`
		GitHub:
		github.com/adityapatel5912/RepoScope
		VS Code / Open VSX:
		adityapatel5912.reposcope

Serves on 127.0.0.1:4100 with zero cloud configuration. [Terminal In-Place Runner]	Writes verified MCP configs for Claude Code, Cursor, OpenCode, Codex & Windsurf. [Universal Multi-Agent MCP]	License: Source-Available License 1.0 (free to inspect, customize & self-host). [Free Source-Available 1.0]
RepoScope v2.8 • One Code Graph.	**Shared Across Humans & AI Coding Agents.**	**100% Local Privacy • Zero Context Waste.**